

Traditional IT vs IoT Solutions



Traditional IT Solutions

- User-driven, request-response model
- Centralized servers and data centers
- Stable networks and continuous power
- Transaction-focused (orders, payments, records)



ERP solution



Web application
like e-commerce

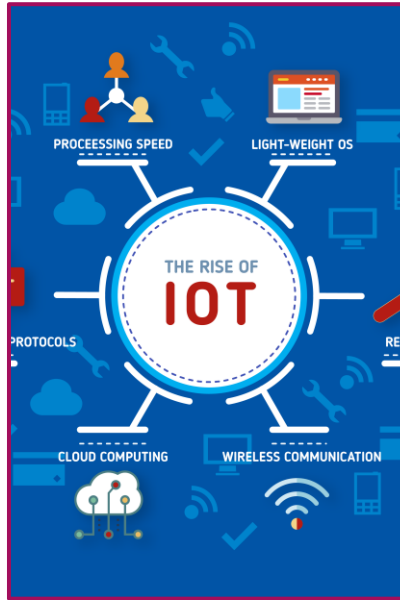


Mobile Applications



Banking, Insurance,
utility Service apps

Traditional IT vs IoT Solutions



IoT Solutions

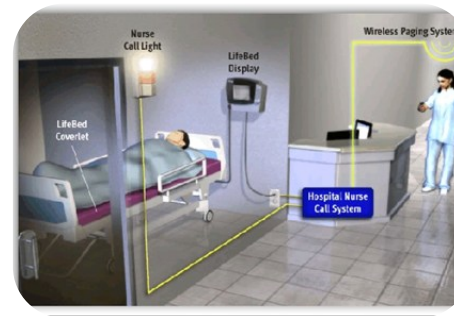
- Event-driven systems (sensor-generated data with Physical world integration)
- Massive scale (thousands/millions of devices)
- Intermittent connectivity, Resource-constrained, battery-powered devices
- Real-time data & automation focus



Smart home
appliance



Smart city'



Smart Hospitals



Military
intelligence

IT VS OT Vs IoT

INFORMATION TECHNOLOGY

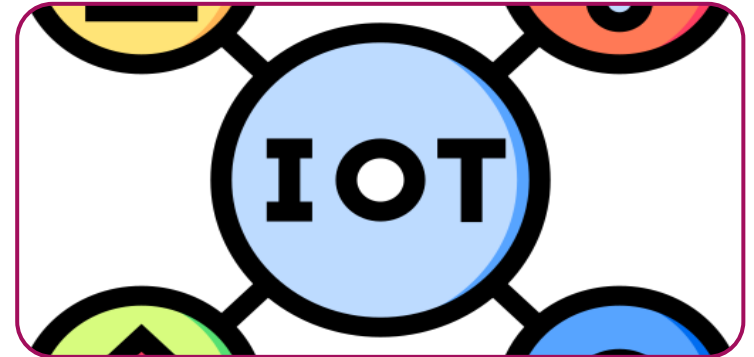
Information Technology (IT)

- ERP, CRM, web apps, databases , applications
- Used in Offices, corporate networks, data centres
- Structured Data, transactions with high Change Tolerance like frequent updates, patches
- Key Priority: Availability, confidentiality, data security



OT (Operational Technology)

- Systems: PLCs, SCADA, DCS, industrial sensors for Physical processes, machines, equipment
- Used mostly in Factories, plants, utilities, field sites
- Real-time control signals, machine states. Low change Tolerance - Stability and safety critical
- Key Priority: Safety, reliability, continuous operation



IoT (Internet of Things)

- Convergence of OT & IT.
- Smart devices, embedded sensors, IoT platforms
- Used in Homes, cities, vehicles, industries — everywhere
- Streaming, event-driven, high-velocity data with moderate
- Change Tolerance — safe, remote, incremental updates
- Key Priority: Intelligence, automation, scalability

IoT vs Traditional IT Systems

Key IoT Design Constraints (vs Traditional IT)

Battery-powered, low-energy devices

Intermittent or unreliable connectivity

Limited CPU, memory, and storage

Harsh physical environments

Massive device scale (thousands to millions)

Device-level security needs

Real-time responsiveness

Long-term device lifecycle management

High-velocity streaming data

Cost-sensitive hardware and operations

IoT vs Traditional IT Systems

